Preparing

```
import os
import io
import base64
import shutil
import random
import string

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from PIL import Image

import tensorflow as tf
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.applications import VGG16, ResNet50
from tensorflow.keras.layers import (
    Conv2D, MaxPooling2D, Flatten, Dense, Dropout, Input
)
from tensorflow.keras.models import Sequential, Model

from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression

import nltk
from nltk.tokenize import word_tokenize

tf.autograph.set_verbosity(0) # Suppress Autograph warnings

print("All libraries imported.")

# Create directories for processed images
os.makedirs('New_Images_Processed', exist_ok=True)
os.makedirs('Damaged_Images_Processed', exist_ok=True)
print('Directories New_Images_Processed and Damaged_Images_Processed created.')

# Helper to decode base64 and save as JPEG
def decode_and_save_image(base64_string, target_dir, filename):
    try:
        if ',' in base64_string:
            _, encoded_data = base64_string.split(',', 1)
        else:
            encoded_data = base64_string

        image_bytes = base64.b64decode(encoded_data)
        image = Image.open(io.BytesIO(image_bytes))
        image = image.convert('RGB')
        file_path = os.path.join(target_dir, filename)
        image.save(file_path, 'JPEG')
    except (AttributeError, ValueError, TypeError, Image.UnidentifiedImageError, FileNotFoundError) as e:
        print(f"Error processing image {filename}: {e}. Skipping this image.")

print('Helper function decode_and_save_image defined.')
```

```
All libraries imported.
Directories New_Images_Processed and Damaged_Images_Processed created.
Helper function decode_and_save_image defined.
```

Data Collection: Loading & Preparation

```
# Load New Image Paths
df_new_images = pd.read_csv('/content/New_images.csv')
new_image_paths = df_new_images['Thumbnail Image']
print('Successfully loaded New_images.csv and extracted new_image_paths.')
print(f'First 5 new image paths:\n{new_image_paths.head()}')
```

```
# Load Damaged Image Paths
df_damaged_images = pd.read_csv('/content/damaged_images.csv')
damaged_image_paths = df_damaged_images['Thumbnail Image']
print('Successfully loaded damaged_images.csv and extracted damaged_image_paths.')
print(f'First 5 damaged image paths:\n{damaged_image_paths.head()}')

# Decode and save all NEW images
for i, base64_str in enumerate(new_image_paths):
    filename = f'new_image_{i}.jpeg'
    try:
        decode_and_save_image(base64_str, 'New_Images_Processed', filename)
    except OSError as e:
        print(f"Error processing image {filename} in loop: {e}. Skipping this image.")
print(f'Successfully processed {len(new_image_paths)} new images from New_Images.csv.')

# Decode and save all DAMAGED images
for i, base64_str in enumerate(damaged_image_paths):
    filename = f'damaged_base64_{i}.jpeg'
    try:
        decode_and_save_image(base64_str, 'Damaged_Images_Processed', filename)
    except OSError as e:
        print(f"Error processing image {filename} in loop: {e}. Skipping this image.")
print(f'Successfully processed {len(damaged_image_paths)} damaged images from damaged_images.csv.')

# Move any existing damaged_image_*.jpeg in /content to processed folder
for filename in os.listdir('/content/'):
    if filename.endswith('.jpeg') and filename.startswith('damaged_image_'):
        shutil.move(os.path.join('/content/', filename), 'Damaged_Images_Processed')
print('Successfully moved any unzipped damaged images to Damaged_Images_Processed directory.')

# Input Path for First Image (New) and Product Name/Description
user_image_path = 'New_Images_Processed/new_image_1.jpeg'
user_text_input_new = 'Laptop Pro 15-inch'
print(f"Using image path for NEW prediction: {user_image_path}")
print(f"Using product name/description (new): {user_text_input_new}")

# Input Path for Second Image (Damaged) and Product Name/Description
user_image_path_2 = 'Damaged_Images_Processed/damaged_base64_35.jpeg'
user_text_input_damaged = (
    'LUCID Memory Foam Soft and Breathable-Machine Washable Mattress Topper Cover, Queen, White'
)
print(f"Using image path for DAMAGED prediction: {user_image_path_2}")
print(f"Using product name/description (damaged): {user_text_input_damaged}")
```

```
Successfully loaded New_images.csv and extracted new_image_paths.
First 5 new image paths:
0    ...
1    ...
2    ...
3    ...
4    ...
Name: Thumbnail Image, dtype: object
Successfully loaded damaged_images.csv and extracted damaged_image_paths.
First 5 damaged image paths:
0    ...
1    ...
2    ...
3    ...
4    ...
Name: Thumbnail Image, dtype: object
Error processing image new_image_30.jpeg in loop: image file is truncated (0 bytes not processed). Skipping this imag
Error processing image new_image_31.jpeg: Incorrect padding. Skipping this image.
Error processing image new_image_58.jpeg in loop: image file is truncated (1 bytes not processed). Skipping this imag
Error processing image new_image_59.jpeg: Incorrect padding. Skipping this image.
Error processing image new_image_66.jpeg: argument of type 'float' is not iterable. Skipping this image.
Error processing image new_image_88.jpeg in loop: image file is truncated (6 bytes not processed). Skipping this imag
Error processing image new_image_89.jpeg: Incorrect padding. Skipping this image.
Error processing image new_image_92.jpeg in loop: image file is truncated (9 bytes not processed). Skipping this imag
Error processing image new_image_93.jpeg: Invalid base64-encoded string: number of data characters (2925) cannot be 1
Successfully processed 104 new images from New_Images.csv.
Error processing image damaged_base64_30.jpeg in loop: image file is truncated (0 bytes not processed). Skipping this
Error processing image damaged_base64_31.jpeg: Incorrect padding. Skipping this image.
Error processing image damaged_base64_58.jpeg in loop: image file is truncated (1 bytes not processed). Skipping this
Error processing image damaged_base64_59.jpeg: Incorrect padding. Skipping this image.
Error processing image damaged_base64_66.jpeg: argument of type 'float' is not iterable. Skipping this image.
Error processing image damaged_base64_88.jpeg in loop: image file is truncated (6 bytes not processed). Skipping this
Error processing image damaged_base64_89.jpeg: Incorrect padding. Skipping this image.
Error processing image damaged_base64_92.jpeg in loop: image file is truncated (9 bytes not processed). Skipping this
Error processing image damaged_base64_93.jpeg: Invalid base64-encoded string: number of data characters (2925) cannot
```

```
Successfully processed 104 damaged images from damaged_images.csv.
Successfully moved any unzipped damaged images to Damaged_Images_Processed directory.
Using image path for NEW prediction: New_Images_Processed/new_image_1.jpeg
Using product name/description (new): Laptop Pro 15-inch
Using image path for DAMAGED prediction: Damaged_Images_Processed/damaged_base64_35.jpeg
Using product name/description (damaged): LUCID Memory Foam Soft and Breathable-Machine Washable Mattress Topper Cove
```

Data Exploration & Visualization

```
class_counts = pd.Series(labels).value_counts().sort_index()

plt.figure(figsize=(6, 5))
class_counts.plot(kind='bar', color=['skyblue', 'lightcoral'])
plt.title('Distribution of New vs. Damaged Images')
plt.xlabel('Image Class')
plt.ylabel('Number of Images')
plt.xticks(ticks=[0, 1], labels=['New (0)', 'Damaged (1)'], rotation=0)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
print("Class Counts:\n", class_counts)
```

```
Class Counts:
 0    95
1    95
Name: count, dtype: int64
```

```
import os
import io
import base64
import shutil
import random
import string

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from PIL import Image

import tensorflow as tf
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.applications import VGG16, ResNet50
from tensorflow.keras.layers import (
    Conv2D, MaxPooling2D, Flatten, Dense, Dropout, Input
)
from tensorflow.keras.models import Sequential, Model
```

```
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression

import nltk
from nltk.tokenize import word_tokenize

tf.autograph.set_verbosity(0) # Suppress Autograph warnings

# Explore and Visualize Data
new_images_count = len(os.listdir('New_Images_Processed'))
damaged_images_count = len(os.listdir('Damaged_Images_Processed'))
print(f"Number of new images: {new_images_count}")
print(f"Number of damaged images: {damaged_images_count}")

new_image_files = [
    os.path.join('New_Images_Processed', f)
    for f in os.listdir('New_Images_Processed') if f.endswith('.jpeg')
]
damaged_image_files = [
    os.path.join('Damaged_Images_Processed', f)
    for f in os.listdir('Damaged_Images_Processed') if f.endswith('.jpeg')
]

sample_size = 4
sample_new_images = random.sample(new_image_files, min(sample_size, len(new_image_files)))
sample_damaged_images = random.sample(damaged_image_files, min(sample_size, len(damaged_image_files)))

plt.figure(figsize=(15, 8))
plt.suptitle('Sample New Images', fontsize=16)
for i, img_path in enumerate(sample_new_images):
    img = Image.open(img_path)
    plt.subplot(1, sample_size, i + 1)
    plt.imshow(img)
    plt.title(f'New Image {i+1}')
    plt.axis('off')
plt.show()

plt.figure(figsize=(15, 8))
plt.suptitle('Sample Damaged Images', fontsize=16)
for i, img_path in enumerate(sample_damaged_images):
    img = Image.open(img_path)
    plt.subplot(1, sample_size, i + 1)
    plt.imshow(img)
    plt.title(f'Damaged Image {i+1}')
    plt.axis('off')
plt.show()
print('Successfully displayed sample images.')

# Split Data and Apply Augmentation
image_paths = []
labels = []

for filename in os.listdir('New_Images_Processed'):
    if filename.endswith(('.jpeg', '.jpg', '.png')):
        image_paths.append(os.path.join('New_Images_Processed', filename))
        labels.append(0)

for filename in os.listdir('Damaged_Images_Processed'):
    if filename.endswith(('.jpeg', '.jpg', '.png')):
        image_paths.append(os.path.join('Damaged_Images_Processed', filename))
        labels.append(1)

# Plotting distribution of New vs. Damaged Images (moved from previous cell)
class_counts = pd.Series(labels).value_counts().sort_index()

plt.figure(figsize=(6, 5))
class_counts.plot(kind='bar', color=['skyblue', 'lightcoral'])
plt.title('Distribution of New vs. Damaged Images')
plt.xlabel('Image Class')
plt.ylabel('Number of Images')
plt.xticks(ticks=[0, 1], labels=['New (0)', 'Damaged (1)'], rotation=0)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```

```
print("Class Counts:\n", class_counts)

print(f"Total images collected: {len(image_paths)}")
print(f"Total labels collected: {len(labels)}")
print(f"First 5 image paths: {image_paths[:5]}")
print(f"First 5 labels: {labels[:5]}")

image_paths = np.array(image_paths)
labels = np.array(labels)

X_train, X_test, y_train, y_test = train_test_split(
    image_paths, labels, test_size=0.3, random_state=42, stratify=labels
)
X_val, X_test, y_val, y_test = train_test_split(
    X_test, y_test, test_size=0.5, random_state=42, stratify=y_test
)

print(f"Training set size: {len(X_train)}")
print(f"Validation set size: {len(X_val)}")
print(f"Test set size: {len(X_test)}")
print("Class distribution in training set:")
print(pd.Series(y_train).value_counts(normalize=True))
print("Class distribution in validation set:")
print(pd.Series(y_val).value_counts(normalize=True))
print("Class distribution in test set:")
print(pd.Series(y_test).value_counts(normalize=True))

TARGET_SIZE = (224, 224)
print(f"Target image size defined as: {TARGET_SIZE}")

train_datagen = ImageDataGenerator(
    rescale=1./255,
    rotation_range=20,
    width_shift_range=0.2,
    height_shift_range=0.2,
    shear_range=0.2,
    zoom_range=0.2,
    horizontal_flip=True,
    fill_mode='nearest'
)

val_test_datagen = ImageDataGenerator(rescale=1./255)
print('ImageDataGenerators defined.')

BATCH_SIZE = 32

train_generator = train_datagen.flow_from_dataframe(
    pd.DataFrame({'filename': X_train, 'class': y_train.astype(str)}),
    x_col='filename',
    y_col='class',
    target_size=TARGET_SIZE,
    batch_size=BATCH_SIZE,
    class_mode='binary',
    seed=42
)

validation_generator = val_test_datagen.flow_from_dataframe(
    pd.DataFrame({'filename': X_val, 'class': y_val.astype(str)}),
    x_col='filename',
    y_col='class',
    target_size=TARGET_SIZE,
    batch_size=BATCH_SIZE,
    class_mode='binary',
    seed=42
)

test_generator = val_test_datagen.flow_from_dataframe(
    pd.DataFrame({'filename': X_test, 'class': y_test.astype(str)}),
    x_col='filename',
    y_col='class',
    target_size=TARGET_SIZE,
    batch_size=BATCH_SIZE,
    class_mode='binary',
    seed=42,
    shuffle=False
)
print('Data generators created.')
```

```
Number of new images: 95
Number of damaged images: 95
```

Sample New Images

Sample Damaged Images

Text Data Processing

```
# Create dummy product_data.csv and load
dummy_data = {
    'product_name': [
        'Laptop Pro 15-inch', 'Smartphone X', 'Wireless Earbuds',
        'Smartwatch 2.0', 'Portable Speaker'
    ]
}
df_dummy = pd.DataFrame(dummy_data)
df_dummy.to_csv('/content/product_data.csv', index=False)

df_text_data = pd.read_csv('/content/product_data.csv')
print('product_data.csv created and loaded.')
print(df_text_data.head())

# Download NLTK resources
try:
    nltk.data.find('tokenizers/punkt')
except LookupError:
    nltk.download('punkt')

try:
    nltk.data.find('tokenizers/punkt_tab/english/')
except LookupError:
    nltk.download('punkt_tab')

def preprocess_text(text):
    if pd.isna(text):
        return []
    text = str(text).lower()
    text = ''.join([c for c in text if c not in string.punctuation])
    tokens = word_tokenize(text)
    return tokens

df_text_data['processed_product_name'] = df_text_data['product_name'].apply(preprocess_text)
print('Basic text preprocessing applied.')
print(df_text_data[['product_name', 'processed_product_name']].head())
```

```
# Synthetic labeled text dataset for text classifier
text_classification_data = {
    'product_name': [
        'new laptop, no damage', 'broken screen, damaged phone',
        'pristine smartwatch', 'scratched earbuds, faulty',
        'brand new portable speaker', 'damaged item, box torn',
        'perfect condition tablet', 'cracked lens camera',
        'new keyboard, sealed', 'damaged mouse, not working'
    ],
    'label': ['new', 'damaged', 'new', 'damaged', 'new',
              'damaged', 'new', 'damaged', 'new', 'damaged']
}
df_text_labels = pd.DataFrame(text_classification_data)

def preprocess_text_for_vectorizer(text):
    if pd.isna(text):
        return ""
    text = str(text).lower()
    text = ''.join([c for c in text if c not in string.punctuation])
    tokens = word_tokenize(text)
    return " ".join(tokens)

df_text_labels['processed_text'] = df_text_labels['product_name'].apply(
    preprocess_text_for_vectorizer
)

X_train_text, X_test_text, y_train_text, y_test_text = train_test_split(
    df_text_labels['processed_text'],
    df_text_labels['label'],
    test_size=0.3,
    random_state=42,
    stratify=df_text_labels['label']
)

tfidf_vectorizer = TfidfVectorizer(max_features=100)
X_train_tfidf = tfidf_vectorizer.fit_transform(X_train_text)
X_test_tfidf = tfidf_vectorizer.transform(X_test_text)

text_model = LogisticRegression(random_state=42)
text_model.fit(X_train_tfidf, y_train_text)

print('Text Classification Model Performance on synthetic data:')
y_pred_text = text_model.predict(X_test_tfidf)
print(classification_report(y_test_text, y_pred_text))

df_text_data['preprocessed_for_model'] = df_text_data['product_name'].apply(
    preprocess_text_for_vectorizer
)
df_text_data_vectorized = tfidf_vectorizer.transform(
    df_text_data['preprocessed_for_model']
)

class_labels = text_model.classes_
prob_damaged_idx = np.where(class_labels == 'damaged')[0][0]
prob_new_idx = np.where(class_labels == 'new')[0][0]

df_text_data['text_prediction_proba_damaged'] = text_model.predict_proba(
    df_text_data_vectorized)[:, prob_damaged_idx]
df_text_data['text_prediction_proba_new'] = text_model.predict_proba(
    df_text_data_vectorized)[:, prob_new_idx]
df_text_data['text_prediction_class'] = text_model.predict(
    df_text_data_vectorized
)

print('Predictions for df_text_data:')
print(df_text_data[['product_name', 'text_prediction_class',
                    'text_prediction_proba_damaged',
                    'text_prediction_proba_new']].head())
print(f"Shape of text embeddings: {df_text_data_vectorized.shape}")
```

```
product_data.csv created and loaded.
         product_name
0  Laptop Pro 15-inch
1        Smartphone X
2    Wireless Earbuds
3      Smartwatch 2.0
4    Portable Speaker
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
Basic text preprocessing applied.
         product_name processed_product_name
0  Laptop Pro 15-inch  [laptop, pro, 15inch]
1        Smartphone X        [smartphone, x]
2    Wireless Earbuds    [wireless, earbuds]
3      Smartwatch 2.0       [smartwatch, 20]
4    Portable Speaker    [portable, speaker]
Text Classification Model Performance on synthetic data:
              precision    recall  f1-score   support

     damaged       1.00      0.50      0.67         2
         new       0.50      1.00      0.67         1

    accuracy                           0.67         3
   macro avg       0.75      0.75      0.67         3
weighted avg       0.83      0.67      0.67         3

Predictions for df_text_data:
         product_name text_prediction_class  text_prediction_proba_damaged  \
0  Laptop Pro 15-inch                   new                       0.382631
1        Smartphone X                   new                       0.424210
2    Wireless Earbuds                   new                       0.490153
3      Smartwatch 2.0                   new                       0.366274
4    Portable Speaker                   new                       0.365863

   text_prediction_proba_new
0                   0.617369
1                   0.575790
2                   0.509847
3                   0.633726
4                   0.634137
Shape of text embeddings: (5, 22)
```

Model Design & Training

- **VGG16 model**
- **ResNet50 Model**
- **Custom CNN model**

```
# Implement and Train VGG16 Model
vgg_base = VGG16(weights='imagenet', include_top=False,
                 input_shape=TARGET_SIZE + (3,))
for layer in vgg_base.layers:
    layer.trainable = False

x = vgg_base.output
x = Flatten()(x)
x = Dense(256, activation='relu')(x)
vgg_predictions = Dense(1, activation='sigmoid')(x)
vgg_model = Model(inputs=vgg_base.input, outputs=vgg_predictions)
vgg_model.compile(
    optimizer='adam', loss='binary_crossentropy',
    metrics=['accuracy'], run_eagerly=True
)
vgg_model.summary()
print('VGG16 model ready.')

# Implement and Train ResNet50 Model
resnet_base = ResNet50(weights='imagenet', include_top=False,
                       input_shape=TARGET_SIZE + (3,))
for layer in resnet_base.layers:
    layer.trainable = False

x_r = resnet_base.output
x_r = Flatten()(x_r)
x_r = Dense(256, activation='relu')(x_r)
resnet_predictions = Dense(1, activation='sigmoid')(x_r)
resnet_model = Model(inputs=resnet_base.input, outputs=resnet_predictions)
resnet_model.compile(optimizer='adam',
                     loss='binary_crossentropy',
                     metrics=['accuracy'])
resnet_model.summary()
print('ResNet50 model ready.')
```

```
# Implement and Train Custom CNN Model
custom_cnn_model = Sequential([
    Input(shape=TARGET_SIZE + (3,)),
    Conv2D(32, (3, 3), activation='relu'),
    MaxPooling2D((2, 2)),
    Conv2D(64, (3, 3), activation='relu'),
    MaxPooling2D((2, 2)),
    Conv2D(128, (3, 3), activation='relu'),
    MaxPooling2D((2, 2)),
    Flatten(),
    Dense(512, activation='relu', name='dense_6'),
    Dropout(0.5),
    Dense(1, activation='sigmoid')
])
custom_cnn_model.compile(optimizer='adam',
                         loss='binary_crossentropy',
                         metrics=['accuracy'])
custom_cnn_model.summary()
print('Custom CNN model ready.')

EPOCHS = 5

print('Training VGG16 model...')
vgg_history = vgg_model.fit(
    train_generator,
    epochs=EPOCHS,
    validation_data=validation_generator,
    verbose=1
)
print('VGG16 training complete.')

print('Training ResNet50 model...')
resnet_history = resnet_model.fit(
    train_generator,
    epochs=EPOCHS,
    validation_data=validation_generator,
    verbose=1
)
print('ResNet50 training complete.')

print('Training Custom CNN model...')
custom_cnn_history = custom_cnn_model.fit(
    train_generator,
    epochs=EPOCHS,
    validation_data=validation_generator,
    verbose=1
)
print('Custom CNN training complete.')
```

Model: "functional_3"

Layer (type)	Output Shape	Param #
input_layer_3 (InputLayer)	(None, 224, 224, 3)	0
block1_conv1 (Conv2D)	(None, 224, 224, 64)	1,792
block1_conv2 (Conv2D)	(None, 224, 224, 64)	36,928
block1_pool (MaxPooling2D)	(None, 112, 112, 64)	0
block2_conv1 (Conv2D)	(None, 112, 112, 128)	73,856
block2_conv2 (Conv2D)	(None, 112, 112, 128)	147,584
block2_pool (MaxPooling2D)	(None, 56, 56, 128)	0
block3_conv1 (Conv2D)	(None, 56, 56, 256)	295,168
block3_conv2 (Conv2D)	(None, 56, 56, 256)	590,080
block3_conv3 (Conv2D)	(None, 56, 56, 256)	590,080
block3_pool (MaxPooling2D)	(None, 28, 28, 256)	0
block4_conv1 (Conv2D)	(None, 28, 28, 512)	1,180,160
block4_conv2 (Conv2D)	(None, 28, 28, 512)	2,359,808
block4_conv3 (Conv2D)	(None, 28, 28, 512)	2,359,808
block4_pool (MaxPooling2D)	(None, 14, 14, 512)	0
block5_conv1 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv2 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv3 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_pool (MaxPooling2D)	(None, 7, 7, 512)	0
flatten_3 (Flatten)	(None, 25088)	0
dense_5 (Dense)	(None, 256)	6,422,784
dense_6 (Dense)	(None, 1)	257

Total params: 21,137,729 (80.63 MB)
Trainable params: 6,423,041 (24.50 MB)
Non-trainable params: 14,714,688 (56.13 MB)
VGG16 model ready.
Model: "functional_4"

Layer (type)	Output Shape	Param #	Connected to
input_layer_4 (InputLayer)	(None, 224, 224, 3)	0	–
conv1_pad (ZeroPadding2D)	(None, 230, 230, 3)	0	input_layer_4[0]…
conv1_conv (Conv2D)	(None, 112, 112, 64)	9,472	conv1_pad[0][0]
conv1_bn (BatchNormalizatio…	(None, 112, 112, 64)	256	conv1_conv[0][0]
conv1_relu (Activation)	(None, 112, 112, 64)	0	conv1_bn[0][0]
pool1_pad (ZeroPadding2D)	(None, 114, 114, 64)	0	conv1_relu[0][0]
pool1_pool (MaxPooling2D)	(None, 56, 56, 64)	0	pool1_pad[0][0]
conv2_block1_1_conv (Conv2D)	(None, 56, 56, 64)	4,160	pool1_pool[0][0]
conv2_block1_1_bn (BatchNormalizatio…	(None, 56, 56, 64)	256	conv2_block1_1_c…
conv2_block1_1_relu (Activation)	(None, 56, 56, 64)	0	conv2_block1_1_b…
conv2_block1_2_conv (Conv2D)	(None, 56, 56, 64)	36,928	conv2_block1_1_r…
conv2_block1_2_bn (BatchNormalizatio…	(None, 56, 56, 64)	256	conv2_block1_2_c…

conv2_block1_2_relu (Activation)	(None, 56, 56, 64)	0	conv2_block1_2_b…
conv2_block1_0_conv (Conv2D)	(None, 56, 56, 256)	16,640	pool1_pool[0][0]
conv2_block1_3_conv (Conv2D)	(None, 56, 56, 256)	16,640	conv2_block1_2_r…
conv2_block1_0_bn (BatchNormalizatio…	(None, 56, 56, 256)	1,024	conv2_block1_0_c…
conv2_block1_3_bn (BatchNormalizatio…	(None, 56, 56, 256)	1,024	conv2_block1_3_c…
conv2_block1_add (Add)	(None, 56, 56, 256)	0	conv2_block1_0_b… conv2_block1_3_b…
conv2_block1_out (Activation)	(None, 56, 56, 256)	0	conv2_block1_add…
conv2_block2_1_conv (Conv2D)	(None, 56, 56, 64)	16,448	conv2_block1_out…
conv2_block2_1_bn (BatchNormalizatio…	(None, 56, 56, 64)	256	conv2_block2_1_c…
conv2_block2_1_relu (Activation)	(None, 56, 56, 64)	0	conv2_block2_1_b…
conv2_block2_2_conv (Conv2D)	(None, 56, 56, 64)	36,928	conv2_block2_1_r…
conv2_block2_2_bn (BatchNormalizatio…	(None, 56, 56, 64)	256	conv2_block2_2_c…
conv2_block2_2_relu (Activation)	(None, 56, 56, 64)	0	conv2_block2_2_b…
conv2_block2_3_conv (Conv2D)	(None, 56, 56, 256)	16,640	conv2_block2_2_r…
conv2_block2_3_bn (BatchNormalizatio…	(None, 56, 56, 256)	1,024	conv2_block2_3_c…
conv2_block2_add (Add)	(None, 56, 56, 256)	0	conv2_block1_out… conv2_block2_3_b…
conv2_block2_out (Activation)	(None, 56, 56, 256)	0	conv2_block2_add…
conv2_block3_1_conv (Conv2D)	(None, 56, 56, 64)	16,448	conv2_block2_out…
conv2_block3_1_bn (BatchNormalizatio…	(None, 56, 56, 64)	256	conv2_block3_1_c…
conv2_block3_1_relu (Activation)	(None, 56, 56, 64)	0	conv2_block3_1_b…
conv2_block3_2_conv (Conv2D)	(None, 56, 56, 64)	36,928	conv2_block3_1_r…
conv2_block3_2_bn (BatchNormalizatio…	(None, 56, 56, 64)	256	conv2_block3_2_c…
conv2_block3_2_relu (Activation)	(None, 56, 56, 64)	0	conv2_block3_2_b…
conv2_block3_3_conv (Conv2D)	(None, 56, 56, 256)	16,640	conv2_block3_2_r…
conv2_block3_3_bn (BatchNormalizatio…	(None, 56, 56, 256)	1,024	conv2_block3_3_c…
conv2_block3_add (Add)	(None, 56, 56, 256)	0	conv2_block2_out… conv2_block3_3_b…
conv2_block3_out (Activation)	(None, 56, 56, 256)	0	conv2_block3_add…
conv3_block1_1_conv (Conv2D)	(None, 28, 28, 128)	32,896	conv2_block3_out…
conv3_block1_1_bn (BatchNormalizatio…	(None, 28, 28, 128)	512	conv3_block1_1_c…
conv3_block1_1_relu (Activation)	(None, 28, 28, 128)	0	conv3_block1_1_b…
conv3_block1_2_conv (Conv2D)	(None, 28, 28, 128)	147,584	conv3_block1_1_r…

(Conv2D)	128)		
conv3_block1_2_bn (BatchNormalizatio…	(None, 28, 28, 128)	512	conv3_block1_2_c…
conv3_block1_2_relu (Activation)	(None, 28, 28, 128)	0	conv3_block1_2_b…
conv3_block1_0_conv (Conv2D)	(None, 28, 28, 512)	131,584	conv2_block3_out…
conv3_block1_3_conv (Conv2D)	(None, 28, 28, 512)	66,048	conv3_block1_2_r…
conv3_block1_0_bn (BatchNormalizatio…	(None, 28, 28, 512)	2,048	conv3_block1_0_c…
conv3_block1_3_bn (BatchNormalizatio…	(None, 28, 28, 512)	2,048	conv3_block1_3_c…
conv3_block1_add (Add)	(None, 28, 28, 512)	0	conv3_block1_0_b… conv3_block1_3_b…
conv3_block1_out (Activation)	(None, 28, 28, 512)	0	conv3_block1_add…
conv3_block2_1_conv (Conv2D)	(None, 28, 28, 128)	65,664	conv3_block1_out…
conv3_block2_1_bn (BatchNormalizatio…	(None, 28, 28, 128)	512	conv3_block2_1_c…
conv3_block2_1_relu (Activation)	(None, 28, 28, 128)	0	conv3_block2_1_b…
conv3_block2_2_conv (Conv2D)	(None, 28, 28, 128)	147,584	conv3_block2_1_r…
conv3_block2_2_bn (BatchNormalizatio…	(None, 28, 28, 128)	512	conv3_block2_2_c…
conv3_block2_2_relu (Activation)	(None, 28, 28, 128)	0	conv3_block2_2_b…
conv3_block2_3_conv (Conv2D)	(None, 28, 28, 512)	66,048	conv3_block2_2_r…
conv3_block2_3_bn (BatchNormalizatio…	(None, 28, 28, 512)	2,048	conv3_block2_3_c…
conv3_block2_add (Add)	(None, 28, 28, 512)	0	conv3_block1_out… conv3_block2_3_b…
conv3_block2_out (Activation)	(None, 28, 28, 512)	0	conv3_block2_add…
conv3_block3_1_conv (Conv2D)	(None, 28, 28, 128)	65,664	conv3_block2_out…
conv3_block3_1_bn (BatchNormalizatio…	(None, 28, 28, 128)	512	conv3_block3_1_c…
conv3_block3_1_relu (Activation)	(None, 28, 28, 128)	0	conv3_block3_1_b…
conv3_block3_2_conv (Conv2D)	(None, 28, 28, 128)	147,584	conv3_block3_1_r…
conv3_block3_2_bn (BatchNormalizatio…	(None, 28, 28, 128)	512	conv3_block3_2_c…
conv3_block3_2_relu (Activation)	(None, 28, 28, 128)	0	conv3_block3_2_b…
conv3_block3_3_conv (Conv2D)	(None, 28, 28, 512)	66,048	conv3_block3_2_r…
conv3_block3_3_bn (BatchNormalizatio…	(None, 28, 28, 512)	2,048	conv3_block3_3_c…
conv3_block3_add (Add)	(None, 28, 28, 512)	0	conv3_block2_out… conv3_block3_3_b…
conv3_block3_out (Activation)	(None, 28, 28, 512)	0	conv3_block3_add…
conv3_block4_1_conv (Conv2D)	(None, 28, 28, 128)	65,664	conv3_block3_out…
conv3_block4_1_bn (BatchNormalizatio…	(None, 28, 28, 128)	512	conv3_block4_1_c…
conv3_block4_1_relu	(None, 28, 28	0	conv3_block4_1_b…

conv3_block4_1_relu (Activation)	(None, 28, 28, 128)	0	conv3_block4_1_b…
conv3_block4_2_conv (Conv2D)	(None, 28, 28, 128)	147,584	conv3_block4_1_r…
conv3_block4_2_bn (BatchNormalizatio…	(None, 28, 28, 128)	512	conv3_block4_2_c…
conv3_block4_2_relu (Activation)	(None, 28, 28, 128)	0	conv3_block4_2_b…
conv3_block4_3_conv (Conv2D)	(None, 28, 28, 512)	66,048	conv3_block4_2_r…
conv3_block4_3_bn (BatchNormalizatio…	(None, 28, 28, 512)	2,048	conv3_block4_3_c…
conv3_block4_add (Add)	(None, 28, 28, 512)	0	conv3_block3_out… conv3_block4_3_b…
conv3_block4_out (Activation)	(None, 28, 28, 512)	0	conv3_block4_add…
conv4_block1_1_conv (Conv2D)	(None, 14, 14, 256)	131,328	conv3_block4_out…
conv4_block1_1_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block1_1_c…
conv4_block1_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block1_1_b…
conv4_block1_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block1_1_r…
conv4_block1_2_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block1_2_c…
conv4_block1_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block1_2_b…
conv4_block1_0_conv (Conv2D)	(None, 14, 14, 1024)	525,312	conv3_block4_out…
conv4_block1_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block1_2_r…
conv4_block1_0_bn (BatchNormalizatio…	(None, 14, 14, 1024)	4,096	conv4_block1_0_c…
conv4_block1_3_bn (BatchNormalizatio…	(None, 14, 14, 1024)	4,096	conv4_block1_3_c…
conv4_block1_add (Add)	(None, 14, 14, 1024)	0	conv4_block1_0_b… conv4_block1_3_b…
conv4_block1_out (Activation)	(None, 14, 14, 1024)	0	conv4_block1_add…
conv4_block2_1_conv (Conv2D)	(None, 14, 14, 256)	262,400	conv4_block1_out…
conv4_block2_1_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block2_1_c…
conv4_block2_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block2_1_b…
conv4_block2_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block2_1_r…
conv4_block2_2_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block2_2_c…
conv4_block2_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block2_2_b…
conv4_block2_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block2_2_r…
conv4_block2_3_bn (BatchNormalizatio…	(None, 14, 14, 1024)	4,096	conv4_block2_3_c…
conv4_block2_add (Add)	(None, 14, 14, 1024)	0	conv4_block1_out… conv4_block2_3_b…
conv4_block2_out (Activation)	(None, 14, 14, 1024)	0	conv4_block2_add…
conv4_block3_1_conv (Conv2D)	(None, 14, 14, 256)	262,400	conv4_block2_out…

conv4_block3_1_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block3_1_c…
conv4_block3_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block3_1_b…
conv4_block3_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block3_1_r…
conv4_block3_2_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block3_2_c…
conv4_block3_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block3_2_b…
conv4_block3_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block3_2_r…
conv4_block3_3_bn (BatchNormalizatio…	(None, 14, 14, 1024)	4,096	conv4_block3_3_c…
conv4_block3_add (Add)	(None, 14, 14, 1024)	0	conv4_block2_out… conv4_block3_3_b…
conv4_block3_out (Activation)	(None, 14, 14, 1024)	0	conv4_block3_add…
conv4_block4_1_conv (Conv2D)	(None, 14, 14, 256)	262,400	conv4_block3_out…
conv4_block4_1_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block4_1_c…
conv4_block4_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block4_1_b…
conv4_block4_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block4_1_r…
conv4_block4_2_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block4_2_c…
conv4_block4_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block4_2_b…
conv4_block4_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block4_2_r…
conv4_block4_3_bn (BatchNormalizatio…	(None, 14, 14, 1024)	4,096	conv4_block4_3_c…
conv4_block4_add (Add)	(None, 14, 14, 1024)	0	conv4_block3_out… conv4_block4_3_b…
conv4_block4_out (Activation)	(None, 14, 14, 1024)	0	conv4_block4_add…
conv4_block5_1_conv (Conv2D)	(None, 14, 14, 256)	262,400	conv4_block4_out…
conv4_block5_1_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block5_1_c…
conv4_block5_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block5_1_b…
conv4_block5_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block5_1_r…
conv4_block5_2_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block5_2_c…
conv4_block5_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block5_2_b…
conv4_block5_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block5_2_r…
conv4_block5_3_bn (BatchNormalizatio…	(None, 14, 14, 1024)	4,096	conv4_block5_3_c…
conv4_block5_add (Add)	(None, 14, 14, 1024)	0	conv4_block4_out… conv4_block5_3_b…
conv4_block5_out (Activation)	(None, 14, 14, 1024)	0	conv4_block5_add…
conv4_block6_1_conv (Conv2D)	(None, 14, 14, 256)	262,400	conv4_block5_out…
conv4_block6_1_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block6_1_c…

(BatchNormalizatio…	256)		
conv4_block6_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block6_1_b…
conv4_block6_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block6_1_r…
conv4_block6_2_bn (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv4_block6_2_c…
conv4_block6_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block6_2_b…
conv4_block6_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block6_2_r…
conv4_block6_3_bn (BatchNormalizatio…	(None, 14, 14, 1024)	4,096	conv4_block6_3_c…
conv4_block6_add (Add)	(None, 14, 14, 1024)	0	conv4_block5_out… conv4_block6_3_b…
conv4_block6_out (Activation)	(None, 14, 14, 1024)	0	conv4_block6_add…
conv5_block1_1_conv (Conv2D)	(None, 7, 7, 512)	524,800	conv4_block6_out…
conv5_block1_1_bn (BatchNormalizatio…	(None, 7, 7, 512)	2,048	conv5_block1_1_c…
conv5_block1_1_relu (Activation)	(None, 7, 7, 512)	0	conv5_block1_1_b…
conv5_block1_2_conv (Conv2D)	(None, 7, 7, 512)	2,359,808	conv5_block1_1_r…
conv5_block1_2_bn (BatchNormalizatio…	(None, 7, 7, 512)	2,048	conv5_block1_2_c…
conv5_block1_2_relu (Activation)	(None, 7, 7, 512)	0	conv5_block1_2_b…
conv5_block1_0_conv (Conv2D)	(None, 7, 7, 2048)	2,099,200	conv4_block6_out…
conv5_block1_3_conv (Conv2D)	(None, 7, 7, 2048)	1,050,624	conv5_block1_2_r…
conv5_block1_0_bn (BatchNormalizatio…	(None, 7, 7, 2048)	8,192	conv5_block1_0_c…
conv5_block1_3_bn (BatchNormalizatio…	(None, 7, 7, 2048)	8,192	conv5_block1_3_c…
conv5_block1_add (Add)	(None, 7, 7, 2048)	0	conv5_block1_0_b… conv5_block1_3_b…
conv5_block1_out (Activation)	(None, 7, 7, 2048)	0	conv5_block1_add…
conv5_block2_1_conv (Conv2D)	(None, 7, 7, 512)	1,049,088	conv5_block1_out…
conv5_block2_1_bn (BatchNormalizatio…	(None, 7, 7, 512)	2,048	conv5_block2_1_c…
conv5_block2_1_relu (Activation)	(None, 7, 7, 512)	0	conv5_block2_1_b…
conv5_block2_2_conv (Conv2D)	(None, 7, 7, 512)	2,359,808	conv5_block2_1_r…
conv5_block2_2_bn (BatchNormalizatio…	(None, 7, 7, 512)	2,048	conv5_block2_2_c…
conv5_block2_2_relu (Activation)	(None, 7, 7, 512)	0	conv5_block2_2_b…
conv5_block2_3_conv (Conv2D)	(None, 7, 7, 2048)	1,050,624	conv5_block2_2_r…
conv5_block2_3_bn (BatchNormalizatio…	(None, 7, 7, 2048)	8,192	conv5_block2_3_c…
conv5_block2_add (Add)	(None, 7, 7, 2048)	0	conv5_block1_out… conv5_block2_3_b…
conv5_block2_out (Activation)	(None, 7, 7, 2048)	0	conv5_block2_add…
conv5_block3_1_conv	(None, 7, 7, 512)	1,049,088	conv5_block2_out…

conv5_block3_1_conv (Conv2D)	(None, 7, 7, 512)	1,049,088	conv5_block2_out…
conv5_block3_1_bn (BatchNormalizatio…	(None, 7, 7, 512)	2,048	conv5_block3_1_c…
conv5_block3_1_relu (Activation)	(None, 7, 7, 512)	0	conv5_block3_1_b…
conv5_block3_2_conv (Conv2D)	(None, 7, 7, 512)	2,359,808	conv5_block3_1_r…
conv5_block3_2_bn (BatchNormalizatio…	(None, 7, 7, 512)	2,048	conv5_block3_2_c…
conv5_block3_2_relu (Activation)	(None, 7, 7, 512)	0	conv5_block3_2_b…
conv5_block3_3_conv (Conv2D)	(None, 7, 7, 2048)	1,050,624	conv5_block3_2_r…
conv5_block3_3_bn (BatchNormalizatio…	(None, 7, 7, 2048)	8,192	conv5_block3_3_c…
conv5_block3_add (Add)	(None, 7, 7, 2048)	0	conv5_block2_out… conv5_block3_3_b…
conv5_block3_out (Activation)	(None, 7, 7, 2048)	0	conv5_block3_add…
flatten_4 (Flatten)	(None, 100352)	0	conv5_block3_out…
dense_7 (Dense)	(None, 256)	25,690,368	flatten_4[0][0]
dense_8 (Dense)	(None, 1)	257	dense_7[0][0]

Total params: 49,278,337 (187.98 MB)
Trainable params: 25,690,625 (98.00 MB)
Non-trainable params: 23,587,712 (89.98 MB)
ResNet50 model ready.
Model: "sequential_1"

Layer (type)	Output Shape	Param #
conv2d_3 (Conv2D)	(None, 222, 222, 32)	896
max_pooling2d_3 (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_4 (Conv2D)	(None, 109, 109, 64)	18,496
max_pooling2d_4 (MaxPooling2D)	(None, 54, 54, 64)	0
conv2d_5 (Conv2D)	(None, 52, 52, 128)	73,856
max_pooling2d_5 (MaxPooling2D)	(None, 26, 26, 128)	0
flatten_5 (Flatten)	(None, 86528)	0
dense_6 (Dense)	(None, 512)	44,302,848
dropout_1 (Dropout)	(None, 512)	0
dense_9 (Dense)	(None, 1)	513

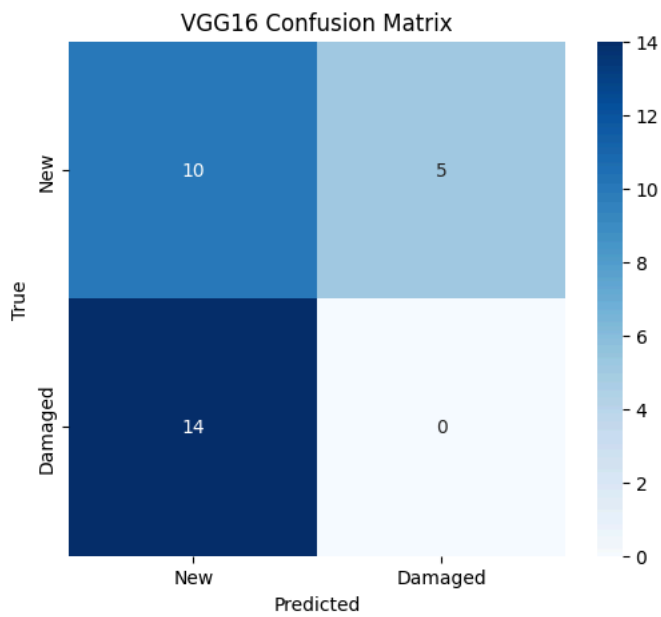
Total params: 44,396,609 (169.36 MB)
Trainable params: 44,396,609 (169.36 MB)
Non-trainable params: 0 (0.00 B)
Custom CNN model ready.
Training VGG16 model...
Epoch 1/5
5/5 ━━━━━━━━━━━━━━━━━━━━ **312s** 59s/step - accuracy: 0.5338 - loss: 4.7492 - val_accuracy: 0.5000 - val_loss: 2.0992
Epoch 2/5
5/5 ━━━━━━━━━━━━━━━━━━━━ **300s** 58s/step - accuracy: 0.5134 - loss: 1.6102 - val_accuracy: 0.4643 - val_loss: 0.8448
Epoch 3/5
5/5 ━━━━━━━━━━━━━━━━━━━━ **305s** 73s/step - accuracy: 0.5131 - loss: 0.9136 - val_accuracy: 0.5357 - val_loss: 1.0829
Epoch 4/5
5/5 ━━━━━━━━━━━━━━━━━━━━ **319s** 58s/step - accuracy: 0.5017 - loss: 0.9649 - val_accuracy: 0.4643 - val_loss: 1.0211
Epoch 5/5
5/5 ━━━━━━━━━━━━━━━━━━━━ **296s** 57s/step - accuracy: 0.5273 - loss: 0.7926 - val_accuracy: 0.5357 - val_loss: 1.2583
VGG16 training complete.
Training ResNet50 model...
Epoch 1/5
5/5 ━━━━━━━━━━━━━━━━━━━━ **46s** 7s/step - accuracy: 0.4821 - loss: 12.5906 - val_accuracy: 0.5000 - val_loss: 11.9481
Epoch 2/5
5/5 ━━━━━━━━━━━━━━━━━━━━ **36s** 6s/step - accuracy: 0.4996 - loss: 9.4693 - val_accuracy: 0.5000 - val_loss: 2.7006
Epoch 3/5
5/5 ━━━━━━━━━━━━━━━━━━━━ **32s** 6s/step - accuracy: 0.5257 - loss: 2.2596 - val_accuracy: 0.5000 - val_loss: 2.2794
Epoch 4/5
5/5 ━━━━━━━━━━━━━━━━━━━━ **32s** 6s/step - accuracy: 0.4735 - loss: 2.2374 - val_accuracy: 0.5000 - val_loss: 1.0187
Epoch 5/5

```
5/5 ━━━━━━━━━━━━━━━━━━━━ 34s 6s/step - accuracy: 0.4420 - loss: 1.1978 - val_accuracy: 0.5929 - val_loss: 0.7914
ResNet50 training complete.
Training Custom CNN model...
Epoch 1/5
5/5 ━━━━━━━━━━━━━━━━━━━━ 24s 4s/step - accuracy: 0.4535 - loss: 3.6284 - val_accuracy: 0.5000 - val_loss: 0.8063
```

Model Evaluation

```
print('Evaluating VGG16 on test set...')
vgg_eval = vgg_model.evaluate(test_generator, verbose=1)
print(f'VGG16 Test Loss: {vgg_eval[0]:.4f}')
print(f'VGG16 Test Accuracy: {vgg_eval[1]:.4f}')

print('Evaluating ResNet50 on test set...')
resnet_eval = resnet_model.evaluate(test_generator, verbose=1)
print(f'ReSNet50 Test Loss: {resnet_eval[0]:.4f}')
print(f'ReSNet50 Test Accuracy: {resnet_eval[1]:.4f}')

print('Evaluating Custom CNN on test set...')
custom_cnn_eval = custom_cnn_model.evaluate(test_generator, verbose=1)
print(f'Custom CNN Test Loss: {custom_cnn_eval[0]:.4f}')
print(f'Custom CNN Test Accuracy: {custom_cnn_eval[1]:.4f}')

print('Computing predictions for classification reports...')
vgg_pred_probs = vgg_model.predict(test_generator)
vgg_pred_labels = (vgg_pred_probs > 0.5).astype(int)

resnet_pred_probs = resnet_model.predict(test_generator)
resnet_pred_labels = (resnet_pred_probs > 0.5).astype(int)

custom_cnn_pred_probs = custom_cnn_model.predict(test_generator)
custom_cnn_pred_labels = (custom_cnn_pred_probs > 0.5).astype(int)

y_true = test_generator.classes

print('\n--- VGG16 Model Performance ---')
print(classification_report(y_true, vgg_pred_labels))
plt.figure(figsize=(6, 5))
sns.heatmap(confusion_matrix(y_true, vgg_pred_labels),
            annot=True, fmt='d', cmap='Blues',
            xticklabels=['New', 'Damaged'],
            yticklabels=['New', 'Damaged'])
plt.title('VGG16 Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('True')
plt.show()

print('\n--- ResNet50 Model Performance ---')
print(classification_report(y_true, resnet_pred_labels))
plt.figure(figsize=(6, 5))
sns.heatmap(confusion_matrix(y_true, resnet_pred_labels),
            annot=True, fmt='d', cmap='Greens',
            xticklabels=['New', 'Damaged'],
            yticklabels=['New', 'Damaged'])
plt.title('ResNet50 Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('True')
plt.show()

print('\n--- Custom CNN Model Performance ---')
print(classification_report(y_true, custom_cnn_pred_labels))
plt.figure(figsize=(6, 5))
sns.heatmap(confusion_matrix(y_true, custom_cnn_pred_labels),
            annot=True, fmt='d', cmap='Oranges',
            xticklabels=['New', 'Damaged'],
            yticklabels=['New', 'Damaged'])
plt.title('Custom CNN Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('True')
plt.show()

print('Classification reports and confusion matrices generated.')
```

```
Evaluating VGG16 on test set...
/usr/local/lib/python3.12/dist-packages/keras/src/trainers/data_adapters/py_dataset_adapter.py:121: UserWarning: Your
  self._warn_if_super_not_called()
1/1 ━━━━━━━━━━━━━━━━━━━━ 23s 23s/step - accuracy: 0.3448 - loss: 1.4701
VGG16 Test Loss: 1.4701
VGG16 Test Accuracy: 0.3448
Evaluating ResNet50 on test set...
1/1 ━━━━━━━━━━━━━━━━━━━━ 5s 5s/step - accuracy: 0.4828 - loss: 0.7365
ReSNet50 Test Loss: 0.7365
ReSNet50 Test Accuracy: 0.4828
Evaluating Custom CNN on test set...
1/1 ━━━━━━━━━━━━━━━━━━━━ 1s 1s/step - accuracy: 0.4828 - loss: 0.6952
Custom CNN Test Loss: 0.6952
Custom CNN Test Accuracy: 0.4828
Computing predictions for classification reports...
1/1 ━━━━━━━━━━━━━━━━━━━━ 20s 20s/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 8s 8s/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 1s 1s/step

--- VGG16 Model Performance ---
              precision    recall  f1-score   support

           0       0.42      0.67      0.51        15
           1       0.00      0.00      0.00        14

    accuracy                           0.34        29
   macro avg       0.21      0.33      0.26        29
weighted avg       0.22      0.34      0.27        29
```




```
--- ResNet50 Model Performance ---
              precision    recall  f1-score   support

           0       0.50      0.67      0.57        15
           1       0.44      0.29      0.35        14
```

Image Feature Extraction (for multimodal part)

```
import requests
from PIL import Image
from io import BytesIO

def preprocess_image(image_path):
    try:
        if image_path.startswith('http://') or image_path.startswith('https://'):
            response = requests.get(image_path)
            response.raise_for_status()  # Raise an HTTPError for bad responses (4xx or 5xx)
            img = Image.open(BytesIO(response.content)).convert('RGB')
        else:
            img = Image.open(image_path).convert('RGB')

        img = img.resize(TARGET_SIZE)
        img_array = np.array(img) / 255.0
        img_array = np.expand_dims(img_array, axis=0)
        return img_array
    except FileNotFoundError:
        print(f"Error: Image file not found at {image_path}")
        return None
```

```
    except requests.exceptions.RequestException as e:
        print(f"Error downloading image from {image_path}: {e}")
        return None
    except Exception as e:
        print(f"Error processing image {image_path}: {e}")
        return None

processed_image_new = preprocess_image(user_image_path)
processed_image = preprocess_image(user_image_path_2)

if processed_image_new is not None:
    print(f"New preprocessed image shape: {processed_image_new.shape}")
if processed_image is not None:
    print(f"Second preprocessed image shape: {processed_image.shape}")

# Predict probabilities for both images using the custom CNN
if processed_image_new is not None:
    prediction_probability_new = custom_cnn_model.predict(processed_image_new)[0][0]
    print(f"Prediction probability for new image (Custom CNN): {prediction_probability_new:.4f}")
else:
    prediction_probability_new = None

if processed_image is not None:
    prediction_probability = custom_cnn_model.predict(processed_image)[0][0]
    print(f"Prediction probability for second image (Custom CNN): {prediction_probability:.4f}")
else:
    prediction_probability = None

# Simple refund eligibility logic for new and damaged images
if processed_image_new is not None and prediction_probability_new is not None:
    THRESHOLD_DAMAGED = 0.8
    THRESHOLD_NEW = 0.66
    if prediction_probability_new >= THRESHOLD_DAMAGED:
        classification_new = 'Damaged'
        refund_eligibility_new = 'Eligible for Refund (High Confidence)'
    elif prediction_probability_new <= THRESHOLD_NEW:
        classification_new = 'New'
        refund_eligibility_new = 'Not Eligible for Refund (High Confidence)'
    else:
        classification_new = 'Uncertain'
        refund_eligibility_new = 'Requires Manual Review'
    print(f"\nImage Classification (New Image): {classification_new}")
    print(f"Refund Eligibility (New Image): {refund_eligibility_new}")

if processed_image is not None and prediction_probability is not None:
    THRESHOLD_DAMAGED_2 = 0.7
    THRESHOLD_NEW_2 = 0.3
    if prediction_probability >= THRESHOLD_DAMAGED_2:
        classification = 'Damaged'
        refund_eligibility = 'Eligible for Refund (High Confidence)'
    elif prediction_probability <= THRESHOLD_NEW_2:
        classification = 'New'
        refund_eligibility = 'Not Eligible for Refund (High Confidence)'
    else:
        classification = 'Uncertain'
        refund_eligibility = 'Requires Manual Review'
    print(f"\nImage Classification (Second Image): {classification}")
    print(f"Refund Eligibility (Second Image): {refund_eligibility}")

    # Extract features from penultimate dense layer of custom CNN
feature_extractor = Model(
    inputs=custom_cnn_model.layers[0].input,
    outputs=custom_cnn_model.get_layer('dense_6').output
)

if processed_image_new is not None:
    image_features_new = feature_extractor.predict(processed_image_new)
    print(f"Shape of image features for new image: {image_features_new.shape}")
else:
    image_features_new = None

if processed_image is not None:
    image_features_damaged = feature_extractor.predict(processed_image)
    print(f"Shape of image features for damaged image: {image_features_damaged.shape}")
else:
    image_features_damaged = None
```

```
# Select first two text feature rows for pairing
text_features_new = df_text_data_vectorized.toarray()[0].reshape(1, -1)
text_features_damaged = df_text_data_vectorized.toarray()[1].reshape(1, -1)
print(f"Shape of text features for new image: {text_features_new.shape}")
print(f"Shape of text features for damaged image: {text_features_damaged.shape}")

if image_features_new is not None:
    combined_features_new = np.concatenate((image_features_new, text_features_new), axis=1)
    print(f"Shape of combined features for new image: {combined_features_new.shape}")
else:
    combined_features_new = None

if image_features_damaged is not None:
    combined_features_damaged = np.concatenate((image_features_damaged, text_features_damaged), axis=1)
    print(f"Shape of combined features for damaged image: {combined_features_damaged.shape}")
else:
    combined_features_damaged = None
```

```
New preprocessed image shape: (1, 224, 224, 3)
Second preprocessed image shape: (1, 224, 224, 3)
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 209ms/step
Prediction probability for new image (Custom CNN): 0.5077
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 114ms/step
Prediction probability for second image (Custom CNN): 0.5070

Image Classification (New Image): New
Refund Eligibility (New Image): Not Eligible for Refund (High Confidence)

Image Classification (Second Image): Uncertain
Refund Eligibility (Second Image): Requires Manual Review
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 169ms/step
Shape of image features for new image: (1, 512)
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 99ms/step
Shape of image features for damaged image: (1, 512)
Shape of text features for new image: (1, 22)
Shape of text features for damaged image: (1, 22)
Shape of combined features for new image: (1, 534)
Shape of combined features for damaged image: (1, 534)
```

Train a Simple Multimodal Classifier (Logistic Regression)

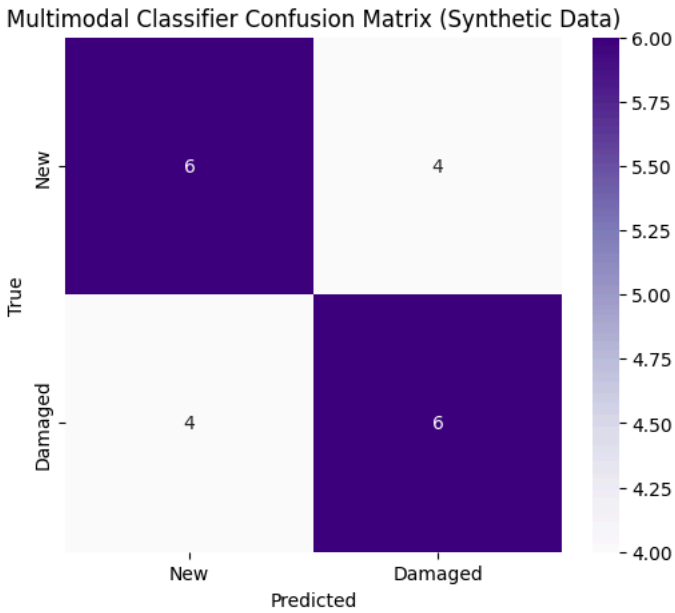
```
num_samples_per_class = 50
num_image_features = 512
num_text_features = 22

synthetic_image_features_new = np.random.rand(num_samples_per_class, num_image_features)
synthetic_text_features_new = np.random.rand(num_samples_per_class, num_text_features)
synthetic_labels_new = np.zeros(num_samples_per_class)

synthetic_image_features_damaged = np.random.rand(num_samples_per_class, num_image_features)
synthetic_text_features_damaged = np.random.rand(num_samples_per_class, num_text_features)
synthetic_labels_damaged = np.ones(num_samples_per_class)

X_synthetic_image = np.concatenate(
    (synthetic_image_features_new, synthetic_image_features_damaged), axis=0
)
X_synthetic_text = np.concatenate(
    (synthetic_text_features_new, synthetic_text_features_damaged), axis=0
)
y_synthetic = np.concatenate(
    (synthetic_labels_new, synthetic_labels_damaged), axis=0
)

X_synthetic_combined = np.concatenate(
    (X_synthetic_image, X_synthetic_text), axis=1
)
print(f"Shape of synthetic combined features: {X_synthetic_combined.shape}")
print(f"Shape of synthetic labels: {y_synthetic.shape}")

X_train_multi, X_test_multi, y_train_multi, y_test_multi = train_test_split(
    X_synthetic_combined, y_synthetic, test_size=0.2,
    random_state=42, stratify=y_synthetic
)

print(f"Shape of multimodal training features: {X_train_multi.shape}")
print(f"Shape of multimodal testing features: {X_test_multi.shape}")
```

```
multimodal_classifier = LogisticRegression(
    random_state=42, max_iter=200
)
multimodal_classifier.fit(X_train_multi, y_train_multi)

y_pred_multi = multimodal_classifier.predict(X_test_multi)
print('\nMultimodal Classifier Performance on synthetic test data:')
print(classification_report(y_test_multi, y_pred_multi))
print(f"Accuracy: {accuracy_score(y_test_multi, y_pred_multi):.4f}")
print('Multimodal classifier trained.')
```

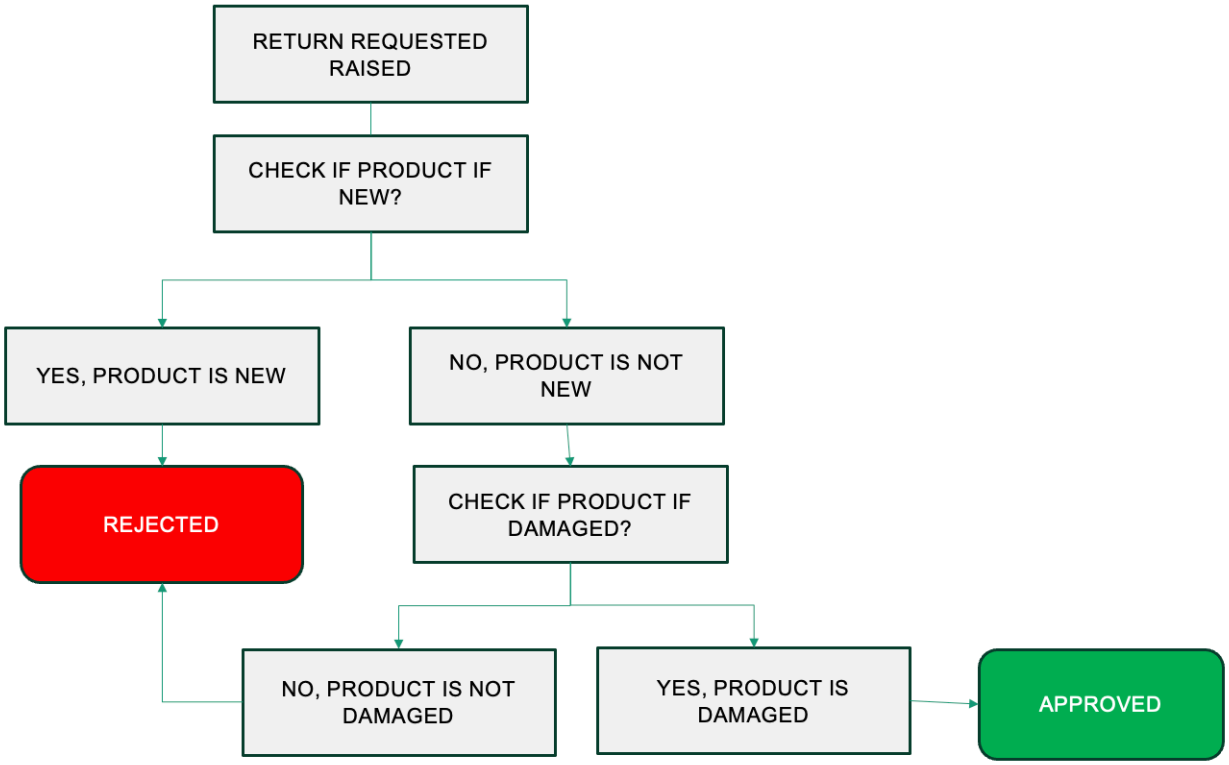
```
Shape of synthetic combined features: (100, 534)
Shape of synthetic labels: (100,)
Shape of multimodal training features: (80, 534)
Shape of multimodal testing features: (20, 534)

Multimodal Classifier Performance on synthetic test data:
              precision    recall  f1-score   support

         0.0       0.60      0.60      0.60        10
         1.0       0.60      0.60      0.60        10

    accuracy                           0.60        20
   macro avg       0.60      0.60      0.60        20
weighted avg       0.60      0.60      0.60        20

Accuracy: 0.6000
Multimodal classifier trained.
```

```
from sklearn.metrics import classification_report, confusion_matrix
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd # Assuming pandas is needed for pd.Series, if not already globally imported

print('\n--- Multimodal Classifier Performance ---')
print(classification_report(y_test_multi, y_pred_multi, zero_division=0))
plt.figure(figsize=(6, 5))
sns.heatmap(confusion_matrix(y_test_multi, y_pred_multi),
            annot=True, fmt='d', cmap='Purples',
            xticklabels=['New', 'Damaged'],
            yticklabels=['New', 'Damaged'])
plt.title('Multimodal Classifier Confusion Matrix (Synthetic Data)')
plt.xlabel('Predicted')
plt.ylabel('True')
plt.show()
```

```
--- Multimodal Classifier Performance ---
              precision    recall  f1-score   support

         0.0       0.60      0.60      0.60        10
         1.0       0.60      0.60      0.60        10

    accuracy                           0.60        20
   macro avg       0.60      0.60      0.60        20
weighted avg       0.60      0.60      0.60        20
```



Prediction – Multimodal (Image + Text)




```
preprocessed_user_text_input = preprocess_text_for_vectorizer(user_text_input_new)
user_text_features_new_input = tfidf_vectorizer.transform(
```

```
    [preprocessed_user_text_input]).toarray()

preprocessed_user_text_input_damaged = preprocess_text_for_vectorizer(
    user_text_input_damaged
)
user_text_features_damaged_input = tfidf_vectorizer.transform(
    [preprocessed_user_text_input_damaged]).toarray()

# Combine CNN image features with text features for user inputs
if image_features_new is not None and user_text_features_new_input is not None:
    combined_features_new_user_input = np.concatenate(
        (image_features_new, user_text_features_new_input), axis=1
    )
else:
    combined_features_new_user_input = None

if image_features_damaged is not None and user_text_features_damaged_input is not None:
    combined_features_damaged_user_input = np.concatenate(
        (image_features_damaged, user_text_features_damaged_input), axis=1
    )
else:
    combined_features_damaged_user_input = None

if combined_features_new_user_input is not None:
    prediction_multimodal_new_user = multimodal_classifier.predict(
        combined_features_new_user_input)[0]
    prediction_proba_multimodal_new_user = multimodal_classifier.predict_proba(
        combined_features_new_user_input)[0]
    predicted_class_new_user = 'New' if prediction_multimodal_new_user == 0 else 'Damaged'
    predicted_proba_damaged_new_user = prediction_proba_multimodal_new_user[1]
```

1. Define the variables for user input

```
# @title # 1. Define the variables for user input
user_image_path = '/content/New_Images_Processed/new_im
user_text_input = 'EUBSWA Camouflage Solar Shower Bag,


print(f"Using image path for prediction: {user_image_pa
print(f"Using product name/description for prediction:
```

user_image_path: "/content/New_Images_Processed/nev"

user_text_input: "EUBSWA Camouflage Solar Shower E"

```
Using image path for prediction: /content/New_Images_Processed/new_image_101.jpeg
Using product name/description for prediction: EUBSWA Camouflage Solar Shower Bag, 5 gallons/20L Solar Heating Campin
```

```
# 2. Preprocess the image
processed_image_prediction = preprocess_image(user_image_path)

# 3. Extract image features
image_features_prediction = None
if processed_image_prediction is not None:
    # Ensure feature_extractor is initialized (it was in Pro7J-e0oMi6, but re-initializing for robustness)
    try:
        _ = feature_extractor
    except NameError:
        feature_extractor = Model(
            inputs=custom_cnn_model.layers[0].input,
            outputs=custom_cnn_model.get_layer('dense_6').output
        )
    image_features_prediction = feature_extractor.predict(processed_image)
    print(f"Shape of image features for prediction: {image_features_prediction.shape}")
else:
    print(f"Image preprocessing failed for {user_image_path}.")

# 4. Preprocess the text
preprocessed_user_text_input = preprocess_text_for_vectorizer(user_text_input)
print(f"Preprocessed text for prediction: {preprocessed_user_text_input}")

# 5. Vectorize the preprocessed text
user_text_features_prediction = tfidf_vectorizer.transform(
    [preprocessed_user_text_input]).toarray()
print(f"Shape of text features for prediction: {user_text_features_prediction.shape}")

# 6. Combine image and text features
combined_features_prediction = None
```

```
if image_features_prediction is not None and user_text_features_prediction is not None:
    combined_features_prediction = np.concatenate(
        (image_features_prediction, user_text_features_prediction), axis=1
    )
    print(f"Shape of combined features for prediction: {combined_features_prediction.shape}")
else:
    print("Cannot combine features: either image or text features are missing.")
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 106ms/step
Shape of image features for prediction: (1, 512)
Preprocessed text for prediction: eubswa camouflage solar shower bag 5 gallons20l solar heating camping shower water
Shape of text features for prediction: (1, 22)
Shape of combined features for prediction: (1, 534)
```

```
if combined_features_prediction is not None:
    # 1. Predict the class and probabilities
    prediction_multimodal_prediction = multimodal_classifier.predict(
        combined_features_prediction
    )[0]
    prediction_proba_multimodal_prediction = multimodal_classifier.predict_proba(
        combined_features_prediction
    )[0]

    # Determine the predicted class label
    predicted_class_prediction = (
        'New' if prediction_multimodal_prediction == 0 else 'Damaged'
    )

    # Extract the probability of the product being 'Damaged'
    predicted_proba_damaged_prediction = prediction_proba_multimodal_prediction[1]

    # 2. Define classification and refund eligibility variables
    classification_prediction = ''
    refund_eligibility_prediction = ''
```